Supplementary Material for WEAVE

Reproducibility, Experimental Setup, and Artifact Ledger

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AAAI 2027 submission supplement

Abstract

This supplement gives the implementation and evaluation details needed to reproduce the experiments reported in the main paper. It consolidates the paper source directory, experiment records, frozen run configurations, training logs, evaluation summaries, and timing records into a path-sanitized description. Machine-specific locations are expressed as variables such as `<REPO_ROOT>`, `<DATA_ROOT>`, `<EXP_ROOT>`, and `<CACHE_ROOT>`; machine-specific paths are not required by the protocol.

1 Scope and Source of Record

The main paper directory is `aaai2027_v4`. The supplement is aligned with the main-paper Table 1, Figure 5, and the method/ablation narrative current as of the AAAI v4 packet. The formal source of record is:

- `aaai2027_v4/paper.tex`: main claims, tables, and captions.
- `aaai2027_v4/fig_data/dino_main.json`: DINO metric sidecar used by the figure scripts.
- `aaai2027_v4/fig_data/method_probes.json`: mechanism-probe sidecar.
- `experiments/architecture/hf_oriented_internal_early_stop.json`: D5 paper training configuration.
- `runs/submission/hf_oriented_internal_early_stop/config.json`: frozen merged run configuration.
- `runs/submission/hf_oriented_internal_early_stop/logs/training_*.csv`: training and internal-probe log.
- `exp/model_probe/inf_timing/generation_only_timing_summary.json`: generation-only timing record.
- `docs/model_probe/target_hf_delta_eval_summary.json`: aggregate target-HF route probe metrics.
- `docs/reproduction/internal_dynamics_early_stop.csv`: compact internal-dynamics curve.

All paths above are relative to `<REPO_ROOT>/WEAVE`. All artifacts are referred to by logical paths so that the supplement is portable across storage layouts.

1.1 Directory Audit and Cleanup Policy

Before writing this supplement, the paper directory and the full experiment tree were audited rather than treated as a single opaque result folder. The audited `exp` tree contained 111 experiment directories, 662 JSON files, 482 CSV files, and 43 log files. The formal supplement only cites records that are needed to reproduce the paper packet. Historical and failed experiments are retained in the ledger as diagnostic context, but they are not used as source-of-truth for the main claims. The source checkout was also audited and found to be an active research worktree rather than a clean release tree: it contained hundreds of uncommitted deletions, modifications, and untracked probe/supplement artifacts. For that reason, this supplement uses committed paper files, frozen run snapshots, and explicit probe summary sidecars as its source of record instead of relying on the ambient working-tree state.

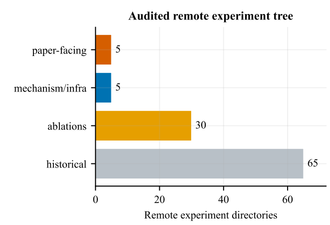

Figure 1: Audited experiment-tree inventory. The paper-facing records are a small subset of a much larger experimental history; the supplement separates source-of-truth runs from diagnostic and historical runs.

Group	Count	Interpretation
Paper-facing runs	5	`dino_s_break`, `main_table`, `bench_all_3060`, `other5_eval`, and `latent_wct_baseline`. These are the directories used for the submission tables, timing, and supporting zero-shot checks.
Mechanism and infra probes	5	`model_probe`, `infra_diag`, `infra_infer_bench`, `infra_opt`, and `param_sweep`. These diagnose speed, target-HF routes, and inference knobs.
Ablation directories	30	`ablation_*`, `abl_*`, `sty_inject`, and `seed3`. These support the causal claims but are not the final checkpoint source.
Historical search runs	65	Earlier `630_*`, `710_*`, `r3_*`, `r4_*`, `hp_*`, `evo_*`, and refactor runs. These explain design convergence and were excluded from the main result ledger unless explicitly referenced.

Cleanup rule. Raw records may contain machine-specific cache paths. The public supplement normalizes all locations into four variables:

Variable	Meaning
<REPO_ROOT>	Checkout containing WEAVE.
<DATA_ROOT>	Dataset and latent-cache root.
<EXP_ROOT>	Experiment output/checkpoint root.
<CACHE_ROOT>	Evaluation feature-cache root.

No absolute paths, host addresses, machine names, or user names are needed to run the protocol.

1.2 Paper Directory Contents

The aaai2027_v4 directory contains the submission source, generated figures, figure sidecars, and build scripts. The 13 files in fig_data are treated as figure/data sidecars rather than raw experiment roots:

File	Size	Role
dino_main.json	11175	DINO-S/DINO-C sidecar for D5-512, P2A-256, and R5-WikiArt.
method_probes.json	16354	Aggregated mechanism-probe source for the method-probe figure.
method_probes_run.log	16356	Reproducibility log for probe aggregation.
method_probe_endpoint.csv	2289	Endpoint/AdaIN ablation values.
method_probe_frequency.csv	3645	Frequency/subband probe values.
method_probe_style_separability.csv	415	Style-separability probe values.
swd_loss_separability.json	1074	SWD-loss diagnostic sidecar.
curve_metrics_hf.csv	19824	Baseline curve metrics used for analysis plots.
teaser_*.png/.jpg	295598	Teaser visual assets for content, WEAVE, SaMam, Seedream, and StyleID.

The compiled paper figures are regenerated by the following scripts: gen_figures.ps1, plot_page1_summary.py, plot_method_probes.py, and make_radar_metric_blocks.py. The supplement does not depend on temporary LaTeX build files such as paper.aux, paper.fls, or prior compile logs.

2 Datasets and Evaluation Boards

D5-512. The primary benchmark is Distinct5-WikiArt at 512 px. It contains five WikiArt styles: Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e. Each style contributes 3600 training images and 150 test images. Evaluation uses all ordered source-target style pairs with 30 content images per source style, giving 750 pairs in total. The 150 same-style pairs are retained as identity calibration rows rather than removed.

P2A-256. The photo-to-art board evaluates transfer at 256 px. It uses the same ordered evaluation logic but with a lower-resolution latent/image cache. It is included to test whether the method remains stable when high-frequency subbands are compressed by resolution.

R5-WikiArt. The broader WikiArt board evaluates style-family generalization. The full evaluation space is all 20 WikiArt families with $20 \times 20 \times 30 = 12000$ ordered pairs. For the cross-method main table, external baselines are reported on the same 750-pair subset, while the WEAVE checkpoint is also used to validate broader family coverage.

Pairing and calibration. All learned methods use the same ordered pair lists within a board. For D5-512 training, source-target pairing uses the frozen pairing cache <DATA_ROOT>/wikiart_distinct5_samam_512_latents_ema/train/.latent_cache/prototype_pairing_top8.pt. The final paper run disables active top-k filtering at training time (pairing_cache_active_topk=0) after loading the cache. Identity rows are used to estimate the no-op style floor and the content ceiling.

2.1 Dataset Layout

A reproducible installation should provide the following logical layout. The names are shown as canonical relative roots; exact physical storage locations are intentionally left variable.

Logical path	Contents
<DATA_ROOT>/wikiart_distinct5_samam_512_latents_ema/train	Packed 512 px latent training cache for the five D5 styles, plus latent-cache metadata.
<DATA_ROOT>/wikiart_distinct5_samam_512_latents_ema/train/.latent_cache/packed	Packed latent tensors consumed by the training dataloader.
<DATA_ROOT>/wikiart_distinct5_samam_512_classview/test	D5-512 held-out class-view image tree used by full evaluation.
<DATA_ROOT>/wikiart_distinct5_512_images/test	D5-512 image tree used as a style-bank root for reference-feature cache generation.
<DATA_ROOT>/wikiarts20_512_test	R5-WikiArt reference/test tree.
<DATA_ROOT>/p2a_256/...	P2A-256 test/cache roots; use the same ordered-pair protocol and metric scripts as D5.

Expected counts. For D5-512, the evaluation board has 5 source styles, 5 target styles, and 30 content images per source style, for 750 ordered evaluations. For P2A-256, identity-style rows are retained in the board definition but some external baselines only provide the 600 off-diagonal transfers; this is why DINO sidecars record both n_images and n_skipped. For R5-WikiArt, the full board is 12000 pairs, while the main table uses the common 750-pair subset for cross-method comparison.

2.2 Data Preprocessing Contract

Training consumes latent tensors rather than raw images. A correct preprocessing run must:

1. Resize/crop images consistently with the target board resolution.

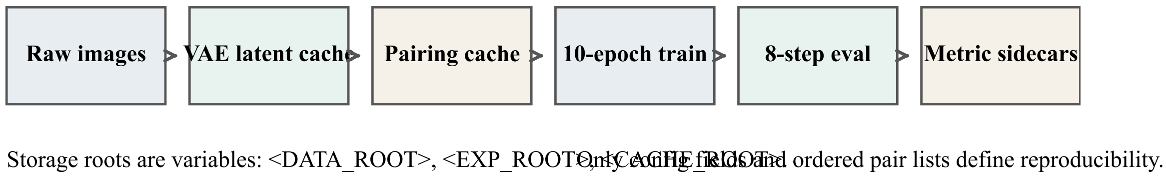

Figure 2: Reproducibility pipeline. The public protocol is defined by dataset artifacts, the frozen config chain, the ordered evaluation boards, and metric sidecars. Physical drive letters and host details are not part of the experiment definition.

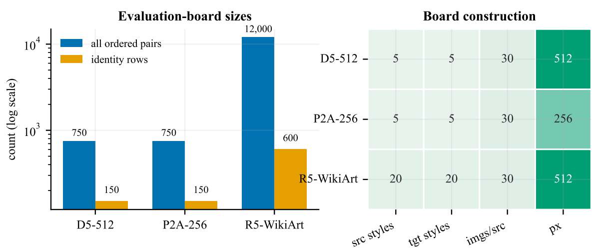

Figure 3: Evaluation-board construction. D5-512 and P2A-256 use 750-pair boards with identity rows retained. R5-WikiArt expands the style-family space to 20 by 20 styles; the main table uses the common 750-pair subset when comparing all methods.

2. Encode images with the SD1.5 EMA VAE and store the latent scaling factor `0.18215` in the config path.
3. Preserve style subdirectories exactly as listed in the config: `Early_Renaissance`, `Impressionism`, `Minimalism`, `Rococo`, and `Ukiyo_e`.
4. Build or provide the prototype pairing cache used by D5 training.
5. Build CLIP/DINO reference caches under `<CACHE_ROOT>` only from held-out test/style-bank images, not from generated outputs.

The public result is reproducible as long as these logical artifacts match. Their physical drive letters and cache locations are not part of the method.

Dataset integrity checks. The dataloader assumes every style directory contains latent tensors encoded with the same VAE, style labels are derived from directory names, and the train/test split is not mixed. A minimal integrity pass should check style directory names, count consistency, latent tensor shape and scale, absence of generated images in reference caches, and whether the pairing cache was built from training latents rather than held-out test images. If a dataset is moved, only root paths should change; style names, cache names, and board construction should remain byte-identical or be regenerated from the same scripts.

3 Hardware and Software Environment

All training and timing numbers reported in the paper come from a single consumer-class GPU. The environment is described by role, not by machine identifier:

Component	Value
GPU class	NVIDIA GeForce RTX 3060, 12 GB
Training precision	bfloat16 automatic mixed precision
TF32 matmul	enabled
Device count	1 GPU

The method itself does not depend on a specific software stack version beyond the ranges listed above; any consumer GPU with ≥ 12 GB VRAM and a working PyTorch CUDA build reproduces the reported numbers.

The software stack follows `WEAVE/requirements.txt`: PyTorch ≥ 2.3, torchvision ≥ 0.18, transformers $4.57 \leq v < 5$, diffusers $0.38 \leq v < 0.39$, LPIPS, torchmetrics, torch-fidelity, Pillow, NumPy, SciPy, scikit-learn, and tqdm. The frozen VAE is the SD1.5 EMA VAE used only for latent encoding/decoding; it contributes 84M frozen parameters and is not trained.

4 Model and Training Setup

WEAVE trains a compact velocity model in SD1.5 VAE latent space. A single-level Haar DWT splits the latent into LL/LH/HL/HH subbands. The base network has four residual blocks of width 64 and three supervised output heads for LL/LH/HL. The D5 model adds separate pooled target-LH and target-HL encoders whose gated residuals enter only the matching velocity heads. The complete model has 1,037,087 trainable parameters; the frozen VAE is reported separately as $1.04\text{M}^{+84\text{M}}$.

Field	Paper configuration
Config	`experiments / architecture / hf_oriented_internal_early_stop.json`
Base config	`config.json`
Seed	42
Maximum epochs	15; internal stop at epoch 4
Batch size	96
Gradient accumulation	1
Optimizer	AdamW, fused when available
Learning rate	2.0×10^{-4}
Minimum learning rate	1.0×10^{-5}
Schedule	cosine
Weight decay	1.0×10^{-4}
Gradient clipping	1.0
Precision	bf16 AMP
Channels-last layout	enabled
Torch compile	disabled
Workers	0
Checkpoint interval	every epoch
Internal probe batch	4 fixed latents at (t=0.5)

The structure-aligned target is constructed in the wavelet basis. The LL target is a mild content-preserving blend,

$$\mathrm{LL}_{target} = 0.7\,\mathrm{LL}_{content} + 0.3\,\mathrm{AdaIN}(\mathrm{LL}_{content}, \mathrm{LL}_{style}),$$

while the high-frequency target uses the target-style LH/HL/HH bands. The implementation loss is the active flow-matching driver. Auxiliary edge and low-pass anchors are retained for compatibility but contribute less than 5 percent of the observed gradient mass in the audited runs.

4.1 Resolved Configuration Details

The frozen merged configuration for the paper run resolves the short public config into the following relevant fields:

4.2 Training Log Audit

The D5 paper run contains four logged epochs, each with 52 optimizer steps and effective batch size 96. The wall-clock epoch time sums to 82.80 s. The probe is evaluated after the optimizer and scheduler step, before checkpoint saving, and does not decode images.

Epoch	Loss	Learning rate	Epoch time (s)	Stop event
1	1.3631	2.0000e-4	25.72	no
2	1.3318	1.9792e-4	19.23	no
3	1.3199	1.9179e-4	18.94	no
4	1.2499	1.8186e-4	18.90	yes

The first epoch is slower because caches and kernels are warmed. Later epochs take about 19 s on a single RTX 3060 (12 GB), including the fixed-latent probe.

4.3 Internal-Dynamics Probe

For a fixed latent minibatch, the implementation performs one backward pass for the LL loss and one for the summed LH/HL loss. Gradients are aggregated over the shared input, attention, AdaLN, and feed-forward modules. The target-HF route and ordinary HF heads are measured separately. The model runs in evaluation mode during the probe, no optimizer step is taken, and `torch.random.fork_rng` restores CPU and CUDA random states.

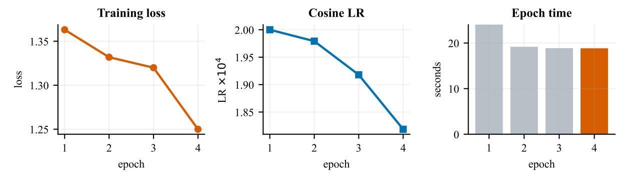

Figure 4: Training audit for the internally selected D5-512 checkpoint. Training stops at epoch 4 when the target-HF gates start expanding and the shared LL/HF gradient ratio first crosses below one.

Epoch	Mean gate	Gate delta	Shared LL/HF	Route/shared HF	Route/HF
1	0.173068	0.000000	1.4727	0.8323	0
2	0.172264	-0.000803	1.4328	0.6133	0
3	0.172009	-0.000255	1.0868	0.4984	0
4	0.175719	+0.003710	0.1634	0.2370	0

Epoch 4 is the first joint event where the gate derivative is positive and the shared LL/HF ratio crosses below one. A retrospective probe of the complete 15-epoch curve selects the same epoch, whose DINO-S is the curve maximum. Residual alignment to the FM target, however, continues to improve after epoch 4 while DINO-S declines. This separates the proposed transition from ordinary loss-based early stopping: it detects a change in route ownership rather than a plateau in endpoint regression.

4.4 Implementation Notes

The implementation is organized around `run.py` for training, `trainer.py` for the optimization loop, `model.py` for the compact velocity network, `wavelet.py` for DWT/IDWT operations, and `utils/run_evaluation.py` for board evaluation. Each training run copies a source snapshot into the run directory; this is why the paper checkpoint can be audited independently of later code cleanup.

Module	Reproducibility role
`run.py`	Loads resolved config, seeds the run, builds dataloader, launches trainer, and optionally invokes full evaluation.
`trainer.py`	Implements optimization, fixed-batch internal probes, checkpoint snapshots, and epoch logs.
`internal_dynamics.py`	Measures route-specific gradients and applies the gate/gradient transition rule.
`model.py`	Defines the WEAVE velocity backbone, oriented target-HF route, heads, solver hooks, and endpoint operations.
`wavelet.py`	Implements Haar DWT/IDWT and subband packing/unpacking used by training and inference.
`utils / run _ evaluation.py`	Generates ordered board outputs, decodes VAE latents, builds/loads caches, computes CLIP/LPIPS, and writes summaries.
`aaai2027_v4/*.py`	Converts sidecars into paper figures and supplement plots.

Field	Value	Relevance
`model.contract_family`	`weave`	Selects the WEAVE architecture contract.
`model.base_dim`	64	Backbone width.
`model.time_dim`	256	Time embedding width.
`model.num_res_blocks`	4	Main residual block count.
`model.num_hires_blocks`	2	High-resolution block count.
`model.num_decoder_blocks`	2	Decoder block count.
`model.target_latent_hf_subband_fusion_enabled`	true	Activates orientation-specific target-HF residuals.
`model.target_latent_hf_head_fusion_gate_init`	0.18	Initial LH/HL route gate.
`model.lowpass_mode`	`dwt_haar`	Single-level Haar decomposition.
`model.endpoint_adain_mode`	`spatial_fiber`	Endpoint style-statistics alignment mode.
`model.style_extrap_alpha`	0.1	Slight target-statistics extrapolation at inference.
`bridge.objective_mode`	`flow_matching`	Rectified-flow training objective.
`bridge.loss_type`	`mse`	Velocity matching loss.
`bridge.t_sampling_mode`	`logit_normal`	Time sampling distribution.
`bridge.bridge_sigma`	0.02	Bridge noise scale.
`bridge.bridge_noise_window`	0.18–0.82	Delayed noise window.
`bridge.training_target_projection_mode`	`dwt`	Target construction in wavelet space.
`bridge.spectral_w_ll/lh/hl/hh`	0.3/1.0/1.0/2.0	Subband loss weighting.
`bridge.ll_partial_alpha`	0.3	Mild LL style-statistics migration.
`training.internal_probe_enabled`	true	Measures fixed-batch loss-path gradients after each epoch.
`training.internal_early_stop_enabled`	true	Stops at the first gate/gradient transition.

Table 1: Resolved fields from the frozen paper configuration.

Implementation details that materially affect reproduction are:

- **Latent scale:** the VAE latent scale factor is `0.18215`.
- **Subband contract:** LL carries structure/color anchor information, while LH/HL/HH carry high-frequency style statistics.
- **Prediction target:** training predicts latent-space velocity/endpoint transport rather than RGB pixels.
- **Output heads:** the compact network supervises LL/LH/HL heads in the audited paper config. HH participates in target/statistics construction and endpoint alignment.
- **Evaluation separation:** generation, VAE decode, CPU image materialization, CLIP/DINO/LPIPS metrics, and CSV writing are timed separately.
- **Cache safety:** reference feature caches are derived from held-out style banks and may be rebuilt; they should not be copied from generated-output folders.

5 Information-Flow Diagnosis

The probe campaign was designed to answer a narrower question than the main table: whether the model's weak style movement came from a wrong target, insufficient training time, or a disconnected conditioning path. The diagnosis favors the third explanation. The structure-aligned target already asks for style in the high-frequency bands:

$$\mathrm{LL}_{target} = 0.7\,\mathrm{LL}_{content} + 0.3\,\mathrm{AdaIN}(\mathrm{LL}_{content}, \mathrm{LL}_{style}),$$
$$\mathrm{LH/HL/HH}_{target} = \mathrm{LH/HL/HH}_{style}.$$

Thus the loss is not merely reconstructing the source image. The failure mode is that, in the baseline, the target image latent is used to construct supervision but does not strongly enter the velocity predictor as an image-specific condition.

Route	Signal path	Probe reading
Baseline style memory	`style_id` → style memory → cross-attention	Coarse style prior. Gates stay near 0.056–0.061, but removing this route hurts full metrics, so it is useful but not sufficient.
Global target-token fusion	target latent → all style tokens → backbone	Connected but too global; it over-controls LL and does not improve useful style transfer.
Raw spatial target-HF	target DWT-HF maps → HF residual maps	Proves capacity exists, but leaks target geometry and collapses content.
Pooled subband target-HF	target DWT-HF → per-subband pooled code → HF residual velocity	Current usable path: stronger style signal without target-coordinate leakage.

The strongest negative control is the raw spatial target-HF route. It raises DINO-S and CLIP-S, which shows that the network can read target-style information when the path is opened. However, the same run reduces DINO-C to 0.4043 and raises LPIPS to 0.5382, so the gain is not a valid style-transfer improvement. It is target-layout leakage. The accepted route therefore compresses target high frequencies into coordinate-free subband codes before they reach the HF velocity heads.

5.1 Probe Protocol and Result Ledger

All probe rows below fine-tune from the same `brk_a_ll03_10ep` checkpoint family and are evaluated on the D5-512 board with the canonical DINOv2-small metrics. The rows are diagnostic architecture probes; they are not mixed into the main table unless the corresponding architecture is fully rerun under the paper protocol.

Run	DINO-S	DINO-C	LPIPS	CLIP-S	Off-DINO-S	Verdict
`target_hf_delta_ft15, ep6`	0.4827	0.7917	0.2950	0.7175	0.3986	Path connected, but modest.
`target_hf_delta_ft15, ep15`	0.4825	0.7913	0.2991	0.7180	0.3987	Longer fine-tune does not solve the bottleneck.
`target_hf_delta_strong_ft6`	0.4870	0.7991	0.2955	0.7176	0.4019	First usable HF route.
`target_hf_spatial_ft6`	0.4901	0.4043	0.5382	0.7483	–	Reject: target geometry leak.
`target_hf_subband_ft6`	**0.4886**	0.7981	0.2966	**0.7209**	0.4039	Primary current architecture probe.
`subband residual ablated`	0.4858	0.7888	0.3010	0.7205	0.4033	Same checkpoint, target-HF residuals zeroed at inference.
`subband residual 1.25x`	0.4873	0.7887	0.3007	0.7201	0.4045	Slight off-style gain but lower balanced metrics.
`subband residual 1.5x`	0.4875	0.7798	0.3054	0.7211	0.4067	Style-biased; content cost too high.
`subband residual HH 1.5x`	0.4878	0.7836	0.3034	0.7206	0.4061	HH-only boost still trades content for off-style.
`target_hf_hybrid_ft6`	0.4858	0.7978	0.2956	0.7197	0.4010	Extra shared/residual mixing does not help.
`target_hf_subband_head_ft6`	0.4873	0.7987	0.2962	0.7192	0.4021	Safe, but below subband residual.
`target_hf_spatial_energy_ft6`	0.4861	0.7909	0.2978	0.7204	0.4027	Content cost without headline gain.
`target_hf_texture_ft6`	0.4860	0.7980	0.2964	0.7182	0.4013	Stationary stats are safe but weak alone.
`target_hf_subband_texture_ft6`	0.4884	**0.7988**	**0.2960**	0.7194	**0.4043**	Conservative alternate with best off-style/content balance.
`target_hf_content_anchor_ft6`	0.4844	0.7955	0.2982	0.7173	0.3995	Safe placement, not competitive.
`target_hf_multitoken_ft6`	0.4836	0.7941	0.2980	0.7187	0.3988	More stationary-stat tokens do not help.
`target_hf_subband_deep_energy_ft6`	0.4826	0.7949	0.2975	0.7176	0.3977	Deeper energy-normalized residual is worse.
`target_hf_subband_film_head_ft6`	0.4826	0.7917	0.2996	0.7180	0.3983	Direct FiLM into HF heads disturbs the pretrained field.
`target_hf_subband_basis_ft6`	0.4828	0.7937	0.2971	0.7183	0.3986	Low-rank content-derived basis is safe but underpowered.
`target_hf_subband_pairstats_ft6`	0.4838	0.7943	0.2971	0.7183	0.3994	Current-target global HF statistics are too coarse.
`target_hf_subband_diraux_ft6`	0.4862	0.7939	0.2974	0.7189	0.4021	Direction auxiliary improves the probe but hurts image metrics.
`subband residual time windows`	0.4866	0.7936	0.2975	0.7194	0.4026	Early/late normalized windows both underperform full residual.
`target_hf_subband_nomem_ft6`	0.4849	0.7948	**0.2943**	0.7167	0.4013	Removing style memory cleans the probe route but weakens style/content metrics.
`target_hf_subband_memres_ft6`	0.4866	0.7935	0.2977	0.7192	0.4025	Explicit style-memory prior subtraction is below subband-only.

Two implementation lessons follow from the probes. First, any promoted target-HF architecture should supervise an HH output path when HH appears in the target. Otherwise the target contains information that no model head can directly predict. Second, more training by itself is not the primary lever: the 15-epoch HF-delta curve remains below the 6-epoch subband route. The useful capacity is frequency-specific and route-specific, not merely total wall-clock exposure.

A matched inference-time ablation further checks that the subband route is causal rather than incidental. Using the trained subband checkpoint, we zero only the three target-HF subband residual modules during evaluation. This lowers DINO-S from 0.4886 to 0.4858, DINO-C from 0.7981 to 0.7888, and worsens LPIPS from 0.2966 to 0.3010. Thus the learned residual path contributes to both style transfer and content-compatible HF transport; it is not a dead branch and not merely a style-over-content shortcut. Scaling the same residual to 1.25x or 1.5x does not improve the balanced point: off-DINO-S and CLIP-S can increase, but DINO-S, DINO-C, and LPIPS worsen. The useful target-HF route is therefore not simply amplitude-limited; the next architecture needs better residual direction or conditioning rather than a scalar gain. This is supported by a direct residual-direction probe: the residual lowers the training-target HF MSE, but its average cosine to the ideal correction direction is only 0.158 and about 98.2% of its energy is orthogonal to that direction. HH is the most aligned band, yet HH-only amplification still follows the same content-cost trend. A direct training auxiliary that aligns the residual to $\mathrm{stopgrad}(v^{\star}_{\mathrm{HF}} - v_{\mathrm{base}})$ increases the probe cosine to 0.322 and the MSE-improvement fraction to 0.117, but lowers DINO-S to 0.4862 and also worsens DINO-C, CLIP-S, and LPIPS. Therefore the direction gap is real, but a separate residual-direction loss is too intrusive in the current transport field. An inference-only time-window probe also rejects the idea that this residual should be moved to only early or only late ODE steps. With normalized half-trajectory windows, early-only and late-only residuals both give DINO-S around 0.4866 and lower DINO-C around 0.7936, below the full residual. Thus the branch is not merely a late texture patch; it should remain a small full-path correction while its basis or conditioning is improved.

A low-rank content-basis variant further tested whether the target-HF code should only select coefficients over residual bases produced from content features. This design disconnects target spatial coordinates, but it lowers DINO-S to 0.4828 and off-DINO-S to 0.3986. The result indicates that the current bottleneck is not solved by making the residual basis more restrictive; the target-HF route still needs enough image-specific orientation and texture capacity.

A pairwise-statistics variant added coordinate-free statistics of the current HF state, the target HF state, and their difference to the per-subband target-HF code. This tests whether the residual needs an explicit "how far from target statistics" signal. The run remains content-safe but lowers DINO-S to 0.4838 and off-DINO-S to 0.3994. Thus global current-target discrepancy descriptors are too coarse for this route; the simple target-only subband code remains the stronger conditioning signal.

The no-memory probe clarifies the role of the style-memory branch. Disabling generic style-memory cross-attention makes the target-HF path easier to read in gradients and aligns auxiliary HF-statistic gradients with the HF-MSE objective, but full evaluation drops DINO-S, DINO-C, CLIP-S, and off-DINO-S relative to the subband route. The visual grid also shows over-smoothed transfers in the Minimalism target column. Thus the remaining design problem is not to delete style memory; it is to keep style memory as a bounded category-level prior while letting target-HF subband codes carry the image-specific residual style.

A follow-up memory-residualized route tested explicit subtraction of a learned style-memory prior from the per-subband target-HF code. It recovered part of the no-memory loss but still trailed the plain subband route on DINO-S, DINO-C, CLIP-S, LPIPS, and off-DINO-S. This negative result argues against algebraically removing the memory prior; future routes should regularize or measure route usage without perturbing the useful category prior.

6 Inference Setup

The main paper uses an 8-step Euler solver. Endpoint high-frequency AdaIN is applied at inference in the wavelet domain. The main D5-512 row uses `runs/submission/hf_oriented_internal_early_stop/epoch_0004.pt`. P2A and R5 use the audited base-WEAVE checkpoints reported before the oriented D5 route was enabled.

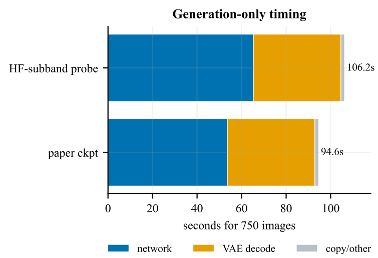

Figure 5: Generation-only timing decomposition on a single RTX 3060 (12 GB). VAE decode is a large fixed cost; the compact velocity network remains the dominant method-specific component.

Inference field	Value
ODE solver	Euler
ODE steps	8
Evaluation batch size	2 for D5-512 generation timing
VAE decode batch size	16 in the timing script
Endpoint AdaIN	enabled
Style extrapolation	$\alpha_{\text{extrap}} = 0.1$
Generated-image count for timing	750
Timing column	generation only, metrics excluded

6.1 Evaluation Runtime Anatomy

The full D5-512 evaluation summary contains both generation and metric timings. A representative full-eval record reports 92.37 s wall time, including 13.52 s WEAVE generation, 2.48 s VAE decode, 6.32 s metric loop, and a large CPU image-copy component when image materialization is enabled. The paper's "Infer" column deliberately uses the separate generation-only timing record, which excludes metric computation and is therefore the correct cross-method speed comparison.

Run	Wall time (s)	Network generation (s)
Base WEAVE	94.63	53.57
Oriented target-HF route	106.25	65.25

6.2 Inference Variants

The following older base-WEAVE variants share one checkpoint and differ only in inference-time endpoint AdaIN strength. They remain diagnostic operating points and are not the updated main D5 result.

The audited generation-only timing on a single RTX 3060 (12 GB) is:

Timing component	Seconds
Total generation-only wall clock	94.63
Velocity-network generation	53.57
VAE decode	39.36
CPU uint8 copy	0.09
Images	750
Wall time per image	126.17 ms
Network time per image	71.42 ms

7 Metrics

DINO-S. DINO-S is the primary style metric. It is the cosine similarity between the DINOv2-small CLS feature of the generated image and the nearest reference feature in the target-style reference pool. The source image is excluded from the reference pool when applicable.

CLIP-S. CLIP-S is the secondary style metric, computed as CLIP ViT-B/32 image-feature cosine similarity to the target-style reference prototype.

DINO-C. DINO-C measures content preservation as DINOv2-small CLS cosine similarity between the generated image and the source image.

LPIPS. LPIPS uses the AlexNet backbone and is reported as a lower-is-better content distance. Radar plots display $1 - \text{LPIPS}$ so all axes point outward for better values.

7.1 IDT–TGT Evaluation Guideline

The guideline adds two source-independent reference outputs to the ordinary metric table. For a request with source image x and target style s, define

$$y_{\text{IDT}}(x, s) = x, \qquad y_{\text{TGT}}(x, s) = r_s, \tag{1}$$

where r_s is a fixed image from the held-out reference pool for style s. The D5 implementation selects the first image in the deterministic reference order and reuses it for every source paired with that target style. Thus TGT depends on the requested style but not on source content. It is an evaluation control, not an input to WEAVE or any baseline.

Why two references are needed. IDT represents a no-op: it preserves all source content but makes no requested edit. A method should therefore score above IDT on DINO-S and CLIP-S before a low LPIPS can be read as successful transfer. TGT represents the opposite extreme: it has the requested style but unrelated content. A content-preserving method should have lower LPIPS and higher DINO-C than TGT. The intuitive acceptable region is therefore bracketed by

$$\begin{gathered} M_{\text{style}}(y) > M_{\text{style}}(y_{\text{IDT}}), \\ \text{LPIPS}(y, x) < \text{LPIPS}(y_{\text{TGT}}, x), \\ \text{DINO-C}(y, x) > \text{DINO-C}(y_{\text{TGT}}, x). \end{gathered} \tag{2}$$

We inspect the inequalities separately and do not combine them into a weighted score.

Interpretation and scope. IDT and TGT are empirical anchors, not mathematical bounds: a generated image may score outside either reference because the metrics are imperfect. Crossing IDT in the wrong direction indicates style suppression under that metric. Moving beyond TGT in content distance identifies a content-loss regime relative to a deliberately unrelated target-style image. This is a prompt

Variant	Scale	DINO-S	CLIP-S	LPIPS	Use
`WEAVE-m`	1.5	0.4843	0.7180	0.2925	Balanced diagnostic point.
`WEAVE-s`	1.6	0.4845	0.7179	0.2867	Peak average of DINO-S and CLIP-S.
`WEAVE-q`	2.0	0.4859	0.7075	0.2583	Previous base-WEAVE max-DINO diagnostic.

Table 2: Inference-only operating points produced from the previous base checkpoint.

for visual inspection, not a verdict on the method outside the current board and metric. On D5-512, IDT is (DINO-S 0.419, CLIP-S 0.693, LPIPS 0, DINO-C 1), while TGT is (1.000, 0.874, 0.733, 0.176). TGT reaches DINO-S 1.000 because the selected reference is part of the same target-style feature pool used by that metric; this value is expected and should not be read as model performance.

7.2 Metric Cache and Determinism

Evaluation uses deterministic seeds and cached reference features. Cache entries are logical artifacts under <CACHE_ROOT>/eval_cache; they may be rebuilt as long as the image roots and model backbones are unchanged. The full-eval summaries record:

- CLIP backend: HuggingFace transformers backend.
- LPIPS network: AlexNet.
- Reference-cache limit per style: 16 for the paper full-eval records.
- Source-feature cache: one cache per board/image-size setting.
- Metric postprocessing: disabled for the main records.

7.3 DINO Sidecar Coverage

The DINO sidecar contains 37 board-method entries. D5-512 and P2A-256 each include 13 methods: Identity, SD-Turbo, StyleAligned, Z-STAR, StyleShot, CUT, SaMST, SaMam, Seedream 4.5, Latent-WCT/WCT, StyleID, AdaIN, and WEAVE as applicable. R5-WikiArt contains 11 entries because not every diagnostic baseline was rerun on the R5 board. The paper table is manually aligned to these sidecars and the CLIP/LPIPS summaries.

Metric interpretation caveats. DINO-S and CLIP-S are style-reference similarities, not human preference labels. DINO-C and LPIPS quantify source preservation, but they penalize different failure modes: LPIPS is sensitive to perceptual pixel/feature distance, whereas DINO-C is a semantic feature similarity. The paper therefore treats the result as a multi-objective operating point. A method with high DINO-S and very high LPIPS is not considered a content-preserving transfer method even if it appears strong on style alone.

8 Main Results Used by the Paper

The main table reports the following WEAVE rows:

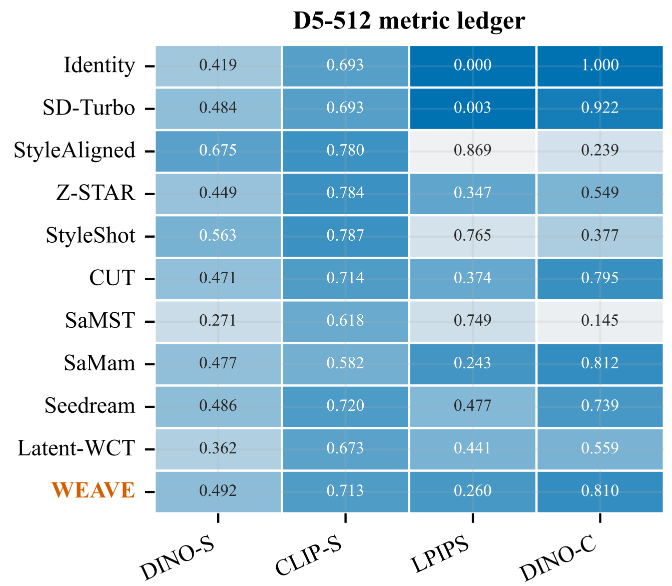

Figure 6: D5-512 metric ledger as a heatmap. LPIPS cells are color-normalized after inversion for readability, while annotations show raw values.

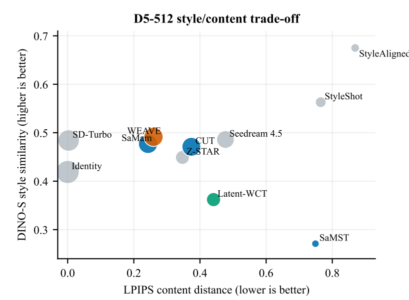

Figure 7: D5-512 style/content trade-off. Marker size grows with DINO-C. WEAVE is not the maximum-style point, but it occupies the low-LPIPS, high-content region among learned methods that clear the identity style floor.

Board	DINO-S	CLIP-S	LPIPS	DINO-C
D5-512 (oriented HF)	0.4915	0.7126	0.2596	0.8103
P2A-256	0.4801	0.6681	0.3116	0.8612
R5-WikiArt	0.5226	0.7747	0.2895	0.7717

The D5-512 claim is a balanced Pareto claim rather than a winner-on-every-axis claim: StyleAligned reaches higher raw DINO-S but with severe content loss, while WEAVE keeps low LPIPS, high DINO-C, about one million trainable parameters, and a 1.4-minute training time on one RTX 3060. The D5 point uses the oriented route; the other boards retain the base model, as stated in the main-table caption.

Table 3: Full automatic-metric ledger used by the main paper. DINO-S, CLIP-S, and DINO-C are higher better; LPIPS is lower better. Dashes mark unavailable or withheld metadata rather than inferred values.

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
	D-S	C-S	LPIPS	D-C	D-S	C-S	LPIPS	D-C	D-S	C-S	LPIPS	D-C	M	min	750 imgs
Identity	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s
Target Style	1.000	0.874	0.733	0.176	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s
SD-Turbo	0.484	0.693	0.003	0.922	0.341	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0^{+1100}	free	5 m
StyleAligned	0.675	0.780	0.869	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829	0.315	0^{+1100}	free	77 m
Z-STAR	0.449	0.784	0.347	0.549	0.514	0.786	0.332	0.526	0.514	0.822	0.384	0.526	0^{+1100}	free	3 h
StyleShot	0.563	0.787	0.765	0.377	0.552	0.774	0.713	0.335	0.484	0.787	0.795	0.380	0^{+2400}	free	5.1 h
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710	0.620	0.795	7.0	322.6	5 m
SaMST	0.271	0.618	0.749	0.145	0.501	0.709	0.398	0.838	0.461	0.667	0.612	0.615	8.3	39.5	10 m
SaMam	0.477	0.582	0.243	0.812	0.505	0.677	0.205	0.870	0.503	0.712	0.227	0.925	8.8	436.0	17.6 m
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	0.227	0.785	0.504	0.742	0.486	0.725	–	API	–
Latent-WCT	0.362	0.673	0.441	0.559	0.338	0.738	0.585	0.386	0.414	0.710	0.444	0.553	0	free	18 s
WEAVE	**0.492**	0.713	0.260	0.810	0.480	0.668	0.312	0.861	0.523	0.775	0.290	0.772	1.04^{+84}	1.4	106 s

8.1 Baseline Accounting

External baselines are separated into three accounting classes:

- **Training-free diffusion/editing baselines:** SD-Turbo, StyleAligned, Z-STAR, StyleShot, Seedream 4.5. Their large frozen backbones are reported with superscript parameter counts in the main table; training is marked as free or API.
- **Learned style-transfer baselines:** CUT, SaMST, SaMam. Their trainable parameter count and training minutes are reported when measured in our runs.
- **Analytic wavelet/statistics baseline:** Latent-WCT. This has zero trained parameters and is included to show that wavelet statistics alone do not explain the WEAVE result.

All methods are evaluated on the same ordered board where outputs are available. Missing or withheld speed/parameter values are marked as unavailable rather than imputed.

8.2 Baseline Data Inventory

The baseline ledger contains both measured baselines and externally sourced or API-limited methods. The evaluation convention is:

1. collect generated images under a method/board/run directory;
2. normalize filenames to the ordered source-target board;
3. evaluate CLIP-S, LPIPS, and optional side metrics using the shared evaluator;
4. evaluate DINO-S and DINO-C using the DINO sidecar scripts;
5. record unavailable timings explicitly rather than estimating them from unrelated hardware.

For D5-512, the paper table includes Identity, SD-Turbo, StyleAligned, Z-STAR, StyleShot, CUT, SaMST, SaMam, Seedream 4.5, Latent-WCT, and WEAVE. For P2A-256 and R5-WikiArt, the same naming convention is used, but some entries have fewer available images or omitted timing fields. This is why each sidecar records image counts and skipped counts rather than assuming all methods have a complete board.

Method group	Output source	Notes
Identity	evaluator no-op	Generated by copying/reading source image as output; defines style floor and content ceiling.
Latent-WCT	analytic transform	Zero trained parameters; evaluates whether wavelet statistics alone are sufficient.
CUT / SaMST / SaMam	learned baselines	Trained or re-evaluated where checkpoints/outputs were available; train/infer times are measured on comparable hardware when reported.
SD-Turbo / StyleAligned / Z-STAR / StyleShot	frozen-backbone editing baselines	Large pretrained backbones; no task-specific training counted, but inference wall-clock is reported when outputs were generated in the evaluation run.
Seedream 4.5	API outputs	Parameter count and internal runtime are withheld; metrics are computed from saved outputs on the shared board.
WEAVE	single-RTX-3060 run	Main checkpoint and timing records come from the audited experiment tree.

9 Ablation and Mechanism Summary

The current evidence supports a compact reading of the method:

$$\begin{aligned} WEAVE = &\text{ Haar wavelet decomposition} \\ &+\text{rectified flow matching} \\ &+\text{endpoint high frequency AdaIN.} \end{aligned}$$

The component ablations and the target-HF probes point to the same causal picture: endpoint statistics and wavelet-domain transport are active, while learned cross-attention memory acts as a coarse prior rather than the main image-specific style injector. This is why the paper frames WEAVE as a compact wavelet-flow model with an auxiliary style memory, not as a transformer-style memory architecture.

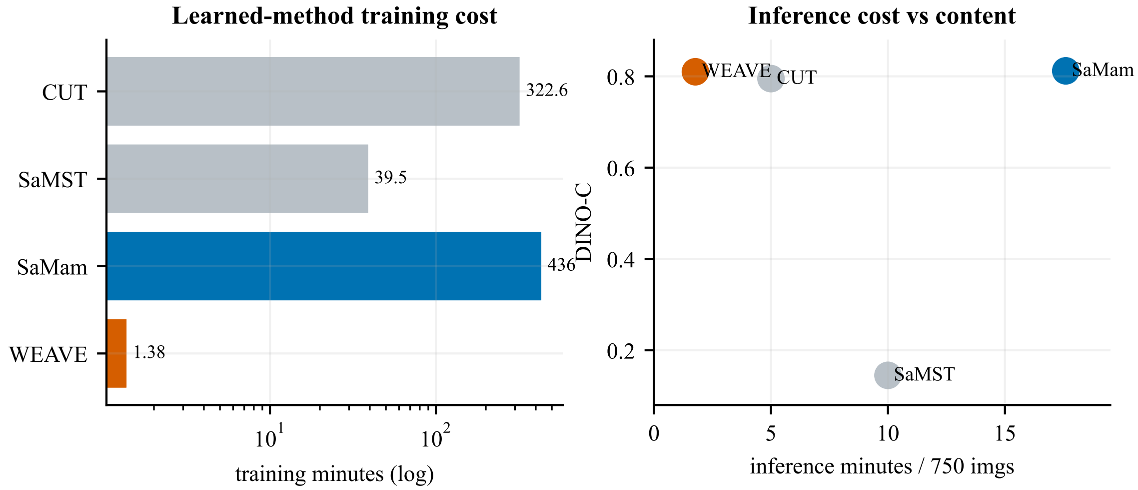

Figure 8: Cost accounting for learned baselines. WEAVE trains in minutes rather than hundreds of minutes and remains in the high-content region at inference.

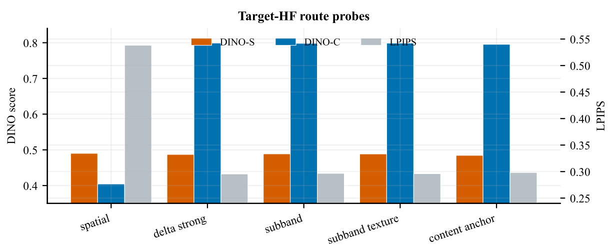

Figure 9: Mechanism-probe comparison. Spatial target-HF injection raises style but collapses content, while pooled/subband target-HF routes preserve content at nearly the same style level.

The cross-attention branch is treated as auxiliary. Probe runs show near-closed gates around 0.056–0.061 and weak gradient mass on the style-memory branch, yet the no-memory subband run underperforms the standard subband route. The effective style movement comes mainly from endpoint AdaIN/statistics, flow transport, and coordinate-free target-HF codes; style memory supplies a useful coarse category prior but cannot replace the image-specific HF route.

9.1 Ablation Directory Summary

The 30 ablation directories are organized into:

- destructive component removals, including no-flow, no-wavelet, no-endpoint-AdaIN, and no-spectral-ODE settings;
- robust/sensitive hyperparameter sweeps, including LL weight, sigma, gate initialization, HH weight, learning rate, loss type, endpoint scale, extrapolation, and ODE step count;
- style-path experiments, including stronger cross-attention, decoder AdaLN/AdaIN, region/global HF injection, and target-HF route variants;
- seed checks for the board metrics.

The consistent outcome is that flow matching, wavelet decomposition, and endpoint high-frequency AdaIN are the active components. Cross-attention and several auxiliary regularizers remain in historical configs for compatibility but are not the causal center of the final result.

10 Next Architecture Plan

The current best probe does not imply that arbitrary style injection is safe. The next architecture step should raise the capacity of the coordinate-free HF route while keeping target spatial coordinates disconnected from the output. The intended route is:

$$\begin{aligned} &x_{style} \to \mathrm{DWT}_{HF} \\ &\to \text{compact per-subband code} \\ &\to \text{orientation-specific HF residual} \\ &\to \text{LH/HL/HH.} \end{aligned}$$

Design choice	Keep / avoid	Reason
Compact per-subband codes	Keep	Current best route; keeps target coordinates disconnected.
Simple target-HF residual ablation evidence	Keep	Zeroing the residual at inference lowers both style and content metrics, confirming causal value.
Orientation-specific residual depth	Keep	LH, HL, and HH carry different stroke/texture directions.
Energy normalization vs. existing HF heads	Keep	Prevents the new route from dominating content structure.
Scalar residual amplification	Avoid	It raises off-style metrics only by spending content budget.
HH-only residual amplification	Avoid	Even the most aligned band does not improve the balanced point when simply scaled.
Direct residual-direction auxiliary	Avoid	It improves the decomposition probe but lowers DINO-S and content metrics, so it competes with the learned transport field.
Time-window residual gating	Avoid	Normalized early/late windows both underperform the full residual, so timing alone is not the bottleneck.
Low-rank content-derived basis	Avoid	It prevents target-coordinate leakage but weakens the target-HF route below the simple subband residual.
Current-target global HF statistics	Avoid	They are train/inference consistent and coordinate-free, but they underperform the target-only subband code.
Stationary-stat multi-token widening	Avoid	A matched probe was worse than subband-only on style and content metrics.
Removing style-memory cross-attention	Avoid	It makes gradient probes cleaner but lowers style/content metrics, so memory should be bounded rather than deleted.
Algebraic memory-prior subtraction	Avoid	It partly recovers from no-memory but remains below the simple subband residual and hurts content.
Raw target HF maps	Avoid	They leak target layout and already failed the content metrics.
LL target-image shortcuts	Avoid	They improve apparent style by spending the content budget.
More gate/epoch tuning first	Avoid	Probe evidence shows route topology, not training length, is the current bottleneck.

Before promoting this route into the main result, the architecture should be rerun on D5-512, P2A-256, and R5-WikiArt with DINO-S as the primary style metric and CLIP-S as a secondary style metric. Any gain that comes with large DINO-C loss or large LPIPS increase should be treated as content collapse rather than a valid style win.

11 Reproduction Commands

The following commands use sanitized variables. They assume the repository has been checked out under <REPO_ROOT> and that the dataset/cache variables point to equivalent storage.

Listing 1: Environment setup

```
cd <REPO_ROOT>/WEAVE
python -m venv .venv
<activate .venv>
pip install torch torchvision --index-url <PYTORCH_CUDA_WHEEL_INDEX
    >
pip install -r requirements.txt
```

Listing 2: Train the D5-512 paper checkpoint

```
cd <REPO_ROOT>/WEAVE
python run.py --config experiments/architecture/
    hf_oriented_internal_early_stop.json
```

Before training on a new machine, replace only storage roots in the config chain:

```
data.data_root = data/train
training.test_image_dir = data/test
training.full_eval_cache_dir = eval_cache
checkpoint.save_dir = runs/submission/
    hf_oriented_internal_early_stop
```

Listing 3: Evaluate a trained checkpoint on D5-512

```
cd <REPO_ROOT>/WEAVE
python scripts/batch_eval_all.py \
  --checkpoint_dir runs/submission/hf_oriented_internal_early_stop
      \
  --config experiments/architecture/hf_oriented_internal_early_stop
      .json \
  --config_override inference.json \
  --epochs 4 --device cuda
```

Listing 4: Evaluate P2A-256 or R5 by swapping roots

```
cd <REPO_ROOT>/WEAVE
python utils/run_evaluation.py \
  <CHECKPOINT>.pt \
  <BOARD_TEST_IMAGE_DIR> \
  <OUTPUT_DIR> \
  --cache_dir <CACHE_ROOT>/eval_cache \
  --clip_hf_cache_dir <CACHE_ROOT>/eval_cache/hf \
  --batch_size <BOARD_BATCH_SIZE> \
  --generation_batch_size <BOARD_BATCH_SIZE> \
  --metric_batch_size <BOARD_BATCH_SIZE> \
  --num_steps 8 \
  --target_chunk_size 1 \
  --vae_decode_batch_size 16 \
  --eval_only_lpips_clip_style \
  --clip_backend hf \
  --eval_lpips_net alex \
  --profile_timing
```

Listing 5: Generation-only timing protocol

```
cd <REPO_ROOT>/WEAVE
python utils/run_evaluation.py \
  <CHECKPOINT>.pt \
  <TEST_IMAGE_DIR> \
  <OUTPUT_DIR> \
  --cache_dir <CACHE_ROOT>/eval_cache \
  --clip_backend none \
  --eval_disable_lpips \
  --batch_size 2 \
  --generation_batch_size 2 \
  --num_steps 8 \
  --target_chunk_size 1 \
  --vae_decode_batch_size 16 \
  --profile_timing \
  --force_regen
```

11.1 End-to-End Checklist

1. Install the Python environment and confirm CUDA sees one RTX 3060-class GPU.
2. Populate <DATA_ROOT> with D5-512 latent training cache and held-out evaluation images.

3. Edit only storage roots in the config chain; keep model, bridge, and training hyperparameters unchanged.
4. Run `python run.py -config experiments/architecture/hf_oriented_internal_early_stop.json`.
5. Verify `epoch_0004.pt`, frozen `config.json`, `internal_dynamics.jsonl`, and training CSV are written.
6. Re-evaluate `epoch_0004.pt` on D5-512 with 8 steps and CLIP/LPIPS enabled.
7. Run the generation-only timing protocol with CLIP and LPIPS disabled.
8. Swap board roots to P2A-256 and R5-WikiArt and rerun the same evaluator.
9. Regenerate figures from `aaai2027_v4/fig_data` and compare table values against `paper.tex`.
10. Confirm no public supplement source contains absolute paths, host endpoints, or usernames.

12 Sanitized Artifact Ledger

12.1 Raw-to-Public Mapping

Raw record class	Public representation	Reason
Checkpoint directories under the experiment tree	`<EXP_ROOT>/<run_name>`	Checkpoints are portable; absolute paths are not.
Dataset image and latent caches	`<DATA_ROOT>/<dataset_name>`	Dataset identity matters; physical storage root does not.
Feature caches	`<CACHE_ROOT>/eval_cache`	Rebuildable from held-out images and metric backbones.
Interactive shell commands and machine identifiers	omitted	They are operational details, not reproducibility requirements.
Temporary audit copies and rendered review PNGs	deleted after verification	They may contain raw paths and are not paper artifacts.

The formal supplement intentionally omits absolute storage paths, host addresses, usernames, and machine identifiers. Those fields exist only in raw run artifacts and are not required for reproducibility once the four storage roots above are supplied.

13 Known Limits

The current supplement does not report a completed human preference study. It also does not claim exhaustive hyperparameter optimization for every heavyweight baseline. The paper's strongest claim is therefore the measured balance among style movement, content preservation, parameter count, and wall-clock cost under the shared automatic protocol, not universal dominance on every metric axis.

Table 4: Sanitized artifact ledger. All paths are logical paths under <REPO_ROOT> or <EXP_ROOT>.

Artifact	Role	Notes
experiments/architecture/hf_oriented_internal_early_stop.json	public config	D5 oriented-HF model and internal stopping rule.
config.json	base config	Dataset board, model defaults, and D5 style list.
runs/submission/hf_oriented_internal_early_stop/config.json	frozen merged config	Run snapshot with all inherited fields resolved.
runs/submission/hf_oriented_internal_early_stop/epoch_0004.pt	checkpoint	Internally selected D5 checkpoint.
runs/submission/hf_oriented_internal_early_stop/logs/training_*.csv	train log	Four epochs, 52 optimizer steps per epoch, and internal-probe columns.
docs/reproduction/internal_dynamics_early_stop.csv	dynamics ledger	Gate and gradient ratios used by the stopping rule.
exp/model_probe/inf_timing/generation_only_timing_summary.json	timing ledger	Oriented-route generation timing for 750 images on RTX 3060.
docs/model_probe/target_hf_delta_eval_summary.json	probe ledger	Aggregate DINO-S, CLIP-S, DINO-C, LPIPS, and off-DINO-S values for target-HF route probes.
docs/713/HANDOFF_2026-07-13.md	handoff note	Human-readable audit, probe diagnosis, run workflow, and next plan.
docs/713/METHOD_EXPLORATION_AND_CKPT_2026-07-13.md	method note	Current checkpoint and method exploration history extracted from older docs.
exp/main_table/p2a_256/full_eval/epoch_0005/summary.json	P2A sidecar	Evaluation summary for the P2A board.
exp/main_table/r5/full_eval/epoch_0005/summary.json	R5 sidecar	Evaluation summary for the R5 board.
aaai2027_v4/fig_data/dino_main.json	figure sidecar	DINO sidecar for paper figure generation and metric audit.
aaai2027_v4/make_radar_metric_blocks.py	figure script	Builds the multi-metric radar figure from table-side data.